

COMRAD Baseline Heatmap at 100M Environment Steps

		IPPO	MAPPO	HAPPO	IDQN	VDN	QMIX	QPLEX-D	QPLEX-Q		
Overall score		0.26	0.24	0.10	0.10	0.10	0.08	0.03	0.06	Overall score	
Stag Hunt Arena		0.49	0.71	0.07	0.40	0.50	0.01	0.04	0.10	Stag Hunt Arena	
Rhythm Sync		0.59	0.65	0.22	0.00	0.00	0.00	0.00	0.00	Rhythm Sync	
Foraging Commons		0.11	0.12	0.10	0.10	0.11	0.12	0.09	0.11	Foraging Commons	
Co-op Puzzle		0.61	0.78	0.45	0.01	0.01	0.00	0.00	0.01	Co-op Puzzle	
Platform Chain		0.04	0.10	0.10	0.01	0.02	0.02	0.07	0.02	Platform Chain	
Armory Siege		0.17	0.20	0.00	0.11	0.18	0.23	0.01	0.14	Armory Siege	
Co-op Health Gathering		0.13	0.07	0.02	0.02	0.07	0.02	0.02	0.03	Co-op Health Gathering	
Lava Pit		0.08	0.09	0.09	0.08	0.07	0.09	0.00	0.09	Lava Pit	
Smart Enemies		0.14	0.17	0.08	0.04	0.06	0.01	0.00	0.01	Smart Enemies	
Dumb Enemies		0.14	0.03	0.01	0.01	0.03	0.01	0.00	0.02	Dumb Enemies	
Ammo Carrier		0.84	0.16	0.16	0.46	0.30	0.28	0.18	0.25	Ammo Carrier	
Stealth Labyrinth		0.00	0.00	0.00	0.00	0.00	0.21	0.00	0.00	Stealth Labyrinth	
Lava Maze		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	Lava Maze	

Scenario-normalized final true_objective scores clipped to [0, 1]. Proxy ceilings: Armory Siege = 100, Lava Maze = 6.